

HAT-P-36b

R-0808

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-36_250323 · created 2026-07-28T11:22:50+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460757.7626 +/- 0.0027 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.131 +/- 0.011
Transit depth (Rp/Rs)^2	1.72 +/- 0.28 [%]
Orbital Inclination (inc)	86.73 +/- 1.67
Airmass coefficient 1 (a1)	1.158 +/- 0.011
Airmass coefficient 2 (a2)	-0.142 +/- 0.028
Scatter in the residuals of the lightcurve fit is	0.92 %
Best Comparison Star	None
Optimal Aperture	3.54
Optimal Annulus	4.89
Transit Duration (day)	0.0945 +/- 0.0029

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.131 $\pm$ 0.011 vs 0.126 (deviation 0.45 σ)
Duration fit vs published	0.0945 d vs 0.0925 d (deviation 0.69 σ)
Mid-time O-C	-0.2 min
Depth SNR	11.9
Residual scatter	0.92 %

Light curve

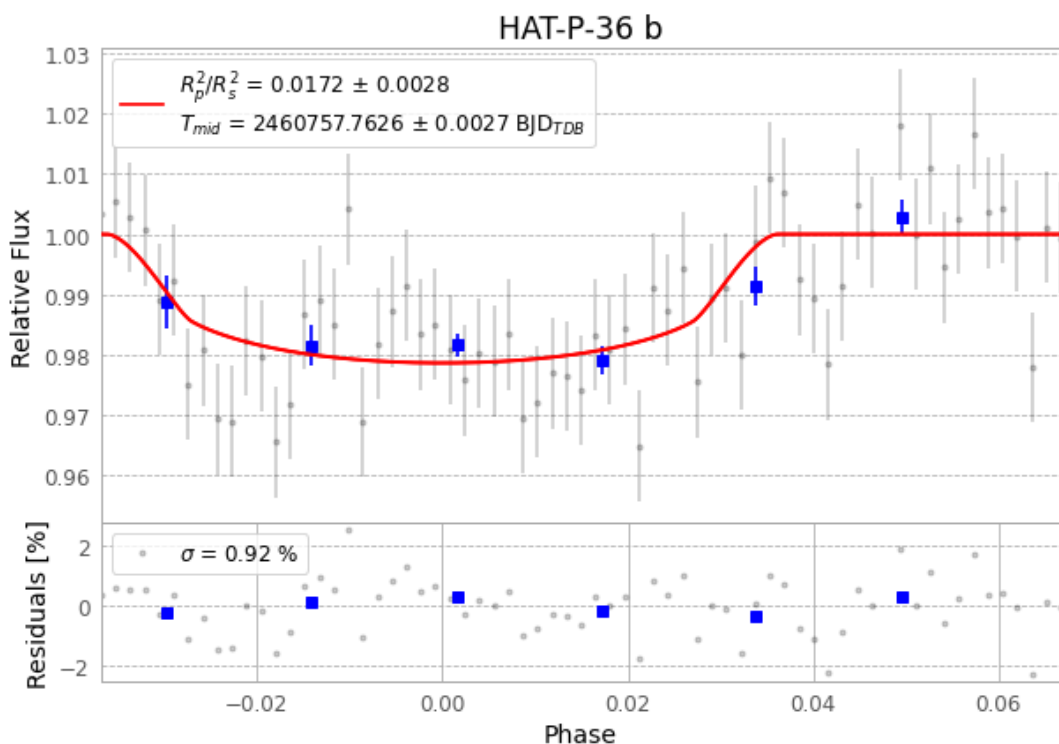


Figure 1 — FinalLightCurve_HAT-P-36 b_2025-03-22.png (SHA-256 331e100fa4a9321b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [389, 226], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 edce7ea9d0af61a5...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

67 FITS frames (44 MB), HATP-36250323050015.FITS → HATP-36250323081815.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-250323.log (2db449252813...), FinalLightCurve_HAT-P-36 b_2025-03-22.png (331e100fa4a9...), FinalLightCurve_HAT-P-36 b_2025-03-22.csv (fb247c362228...)

Compute resources

Wall time 110.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 11:22Z.